HENSELIZATION AND STRICT HENSELIZATION DO NOT CHANGE THE DIMENSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Henselization and strict henselization, with faithful flatness of the maps from the base. the author cites Lemma lemma-dumb-properties-henselization at proof lines 11137-11139. Provides the faithful flatness used for one inequality.
- Dimension and going down for faithfully flat maps. the author cites Algebra, Lemma algebra-lemma-dimension-going-up at proof line 11139. Gives the inequality in the easy direction.
- Dimension is unchanged by etale extensions, and the strict henselization is a directed colimit of localisations of etale algebras. the author cites Lemma lemma-dimension-etale-extension at proof lines 11152-11155. Supplies the equality for each stage of the colimit, which bounds the dimension from above.

Source reference. Original lemma 06LK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8469–8469. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07QM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9837–9837. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00OH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 895–895. Navigation pointer to a statement the

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Source reference. Original lemma 07QP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9839–9839. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Given any local ring R we have* $\dim(R) = \dim(R^h) = \dim(R^{sh})$.

Proof. Since $R \to R^{sh}$ is faithfully flat (Lemma 07QM) we see that $\dim(R^{sh}) \geq \dim(R)$ by going down, see Algebra, Lemma 00OH. For the converse, we write $R^{sh} = \operatorname{colim} R_i$ as a directed colimit of local rings R_i each of which is a localization of an étale R-algebra. Now if $\mathfrak{q}_0 \subset \mathfrak{q}_1 \subset \ldots \subset \mathfrak{q}_n$ is a chain of prime ideals in R^{sh}, then for some sufficiently large i the sequence

$$R_i \cap \mathfrak{q}_0 \subset R_i \cap \mathfrak{q}_1 \subset \ldots \subset R_i \cap \mathfrak{q}_n$$

is a chain of primes in R_i. Thus we see that $\dim(R^{sh}) \leq \sup_i \dim(R_i)$. But by the result of Lemma 07QP we have $\dim(R_i) = \dim(R)$ for each i and we win. $\square$